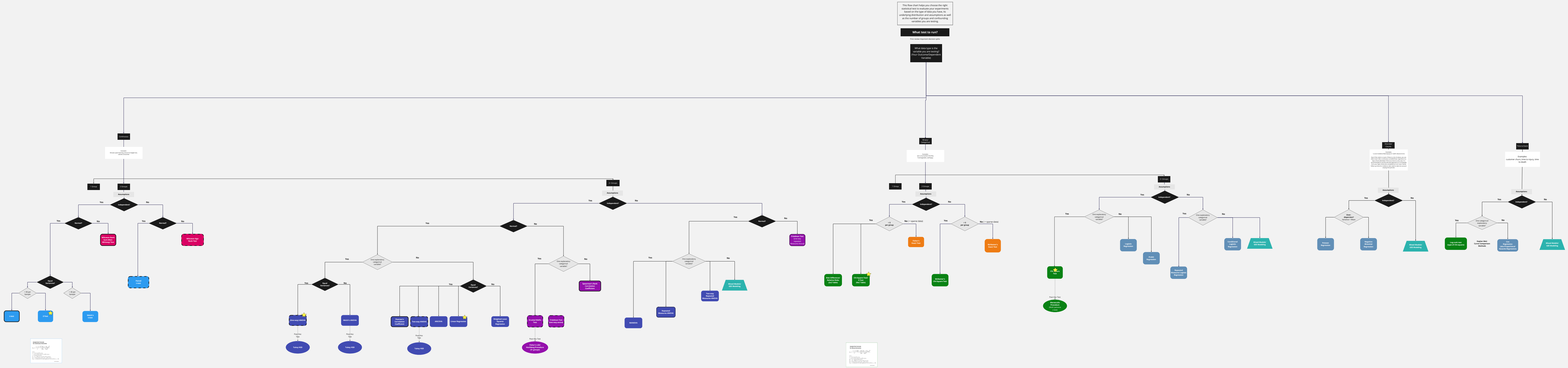


This flow chart helps you choose the right statistical test to evaluate your experiments based on the type of data you have, its underlying distribution and assumptions as well as the number of groups and confounding variables you are testing.

## What test to run?

First review important decision splits

What data type is the variable you are testing? (Your Outcome/Dependent Variable)

**Continuous**

Examples:
Minutes spent learning, amount of weight lost, calories consumed

- 1 Group
- 2 Groups
  - Assumptions
  - Independent?
    - Yes → Normal?
      - Yes → Equal Variances?
        - < 30 per Sample → t-test; Z-Test
        - > 30 per Sample → Welch's t-test
      - No → Wilcoxon Rank Sum (Man-Whitney) Test
    - No → Normal?
      - Yes → Paired t-test
      - No → Wilcoxon Sign Rank Test
- 2+ Groups
  - Assumptions
  - Independent?
    - Yes → Normal?
      - Yes → One explanatory categorical variable?
        - Yes → Equal Variances?
          - Yes → (One-way) ANOVA → Post Hoc Test → Tukey HSD
          - No → Welch's ANOVA → Post Hoc Test → Tukey HSD
        - No → Equal Variances?
          - Yes → Pearson's Correlation Coefficient; Two-way ANOVA (→ Post Hoc Test → Tukey HSD); ANCOVA; Linear Regression
          - No → Weighted Least Squares Regression
      - No → One explanatory categorical variable?
        - Yes → Kruskal-Wallis Test → Post Hoc Test → Fisher's LSD/ Hochberg Procedure (3+ groups); Friedman Test (two way anova)
        - No → Spearman's Rank Correlation Coefficient
    - No → Normal?
      - Yes → One explanatory categorical variable?
        - Yes → MANOVA; Repeated Measures ANOVA
        - No → Two-way Repeated Measures ANOVA; Mixed Models/ GEE Modeling
      - No → Friedman Test (one-way repeated Measures Anova)

Sample Size Formula for Continuous Outcomes

$$n_1 = \frac{r+1}{r}\frac{\sigma^2 (Z_\beta + Z_{\alpha/2})^2}{(\bar{x}_1 - \bar{x}_2)^2}$$

**Binary/ Categorical**

Examples:
won a tournament (Yes/No), fruit/vegetable, sad/happy

- 1 Group
- 2 Groups
  - Assumptions
  - Independent?
    - Yes → > 5 per group
      - Yes → Risk Difference/ Relative Risks (2x2 Table); Chi-Square Test/ Z-Test (RxC Table)
      - No ( = sparse data) → Fisher's Exact Test
    - No → > 5 per group
      - Yes → McNamer's Chi-Square Test
      - No ( = sparse data) → McNamer's Exact Test
- 2+ Groups
  - Assumptions
  - Independent?
    - Yes → One explanatory categorical variable?
      - Yes → Chi Square Test → Post Hoc Test → Marascuilo Procedure (Multi pairwise z-test)
      - No → Logistic Regression; Probit Regression
    - No → One explanatory categorical variable?
      - Yes → Repeated Measures Logistic Regression
      - No → Conditional Logistic Regression; Mixed Models/ GEE Modeling

Sample Size Formula for Binary Outcomes

$$n_1 = \frac{r+1}{r}\frac{\bar{p}(1-\bar{p})(Z_\beta + Z_{\alpha/2})^2}{(p_1 - p_2)^2}$$

**Discrete/ Counts**

Examples:
1,2,3,4; Incidence Rate (injury); 0,1 (with robust errors)

Even if the metric is count, if there is a lot of values, you can often treat it like a continuous variable/linear regression (i.e. days school attended). If the occurrence is very rare, it is recommended to use this discrete approach (i.e. # of people from your high school who completed an iron man). Varied follow up times for incidence rates need to take into account varying time periods.

- Assumptions
- Independent?
  - Yes → Over-dispersion? Variance > Mean
    - Poisson Regression
    - Negative Binomial Regression
  - No → Mixed Models/ GEE Modeling

**Time to Event**

Examples:
customer churn, time to injury, time to death

- Assumptions
- Independent?
  - Yes → One categorical explanatory variable?
    - Yes → Log-rank test (type of Chi-square)
    - Kaplan Meir Curve Comparison Methods
    - No → Cox Regression (aka Proportional Hazards Regression)
  - No → Mixed Models/ GEE Modeling

### Z-test

**z-test vs t-test**
when you don't know the population variance, use t-test. However, as N gets bigger, t converges to z.

**Pooled vs Un pooled**
- Use pooled variance if the groups are the same size, in addition to the same variances
- Otherwise use un pooled variance also known as the Satterthwaite Approximation (take into account both unequal sample sizes & unequal variances)

### Wilcoxon Rank Sum Test

**General:** A small number of people did X and another independent set of people did Y instead, was there a difference and was it significant?

**Example:** The weight of a group of 11 people who smoked was compared to a group of 11 people who did not smoke. Was their weight different? (Small sample size & unequal distributions of weight between groups - the smokers group had a less variability in weight)

**Solution:** Order the data in each group from "worst" to "best" (greatest loss of weight). Rank each value against ALL other values. Sum the ranks (not the values) within each group and then compare.

**Equivalent (conditional):** *t-test (Parametric)*

### Wilcoxon Sign Rank Test

**General:** A small number of people did X, was there a difference before and after X (pre/post)?

**Example:** Was there a significant difference of weight change after smoking?

**Solution:** Order the data in each group from lowest to highest according their absolute value (ignoring their sign) and excluding 0's. Ties split the rank (values 0,1, 4, 4, 6, 7, 7 = _ 1, 2.5, 2.5, 4, 5.5, 5.5). Next, sum of the ranks in the positive group and the negative group and then compare.

**Equivalent (conditional):** *Paired t-test (Parametric)*

### General Linear Models

Logistic, Poisson, Probit, Negative Binomial regression
**R Example:**
glm(y~x, family=poisson)

1. The residuals (aka errors) are normally distributed.
2. The model parameters-regression coefficients, means, residual variance-are estimated using a technique called Ordinary Least Squares.

Additionally:
- Special case of the Generalized Linear Model
- Response variables need to be continuous

[Source]

### One-way vs Two-way ANOVA vs ANCOVA

In a one-way ANOVA, the one factor or independent variable analyzed has three or more categorical groups. A two-way ANOVA instead compares multiple groups of two factors. An ANCOVA includes both factor and continuous independent variables (covariate).

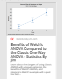
**Example:**
(One-way) ANOVA: Do walruses weight significantly fluctuate every month ?

Two-way ANOVA: Are walruses heavier in early or late mating season and does that depend on the gender of the walrus?

ANCOVA: Do our walruses weigh more in early or late mating season controlling for temperature?

[Walrus source] [Ancova source]

### Welch's Anova



statisticsbyjim.com

Benefits of Welch's ANOVA Compared to the Classic One-Way ANOVA - Statistics By Jim

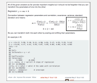
Learn about the dangers of using Classic ANOVA with unequal variances, the benefits of Welch's ANOVA, and interpret a Welch's example with a post hoc test....

### Pearson's Correlation Coefficient vs Linear Regression

Directly mathematically related. The main difference is that you have to pick a "direction" of relatedness for the linear regression.

"In the single predictor case of linear regression, the standardized slope has the same value as the correlation coefficient. "
"The square of Pearson's correlation coefficient is the same as the R^2 in simple linear regression"
[source Stack Exchange Thread]

**General Rule of Thumb:**
0.7? Strong Correlation
0.3> Medium Correlation
0.1> Weak Correlation



All of the given answers so far provide important insights but it should not be forgotten that you can transform the parameters of one into the other:

Regression: $y = mx + b$

Connection between regression parameters and correlation, covariance, variance, standard deviation and means:

$$m = \frac{Cov(y,x)}{Var(x)} = \frac{Cor(y,x) \cdot Sd(y)}{Sd(x)}$$

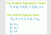
$$b = \bar{y} - m\bar{x}$$

So you can transform both into each other by scaling and shifting their parameters.

An example in R:

### Linear Regression vs ANOVA

These are essentially identical, one possible reason to use ANOVA is to test the similarity of a subset of dummy variables as a whole (but you could also do this using linear models). [Example]



The Multiple Regression Model:
$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$

The ANOVA Special Case:
$Y_{ijk} = \mu + \alpha_j + \beta_k + \epsilon_{ijk}$
$\beta_0 = \mu$
$\beta_1 = \alpha$
$\beta_2 = \beta$

[^ read more here]

### Kruskal-Wallis Test

**General:** 3 or more groups with unequal sizes and/or different variances got different treatments.

**Example:** A small group of users got a package, another big group of users got personalized mail and the third group did not receive anything. Were the engagement levels significantly different between the groups.

***Equivalent (conditional):*** *Anova (parametric), Wilcoxon Rank Sum Test (non-parametric 2 groups only)*

### MANOVA

"Multivariate analysis of variance (MANOVA) is simply an ANOVA with several dependent variables. That is to say, ANOVA tests for the difference in means between two or more groups, while MANOVA tests for the difference in two or more vectors of means." [source] MANOVA also determines the interactions taking place amongst dependent variables.

### Repeated Measures Anova

**General:** Measured X quantity at baseline AND at 2 or more other time points for that same group (if there are only 2 time points, use paired ttest).

**Example:** A group of users got a package. What was the baseline engagement of the group and how did it change over time. Could also compare two independent groups, one that got packages and ones that did not (two-way repeated measures anova).

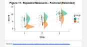
***Equivalent (conditional):*** *McNamer's Chi-Square Test (Binary), Friedman Test (non-parametric)*

### Two-way Repeated Measures Anova

"A two-way repeated measures ANOVA is often used in studies where you have measured a dependent variable over two or more time points, or when subjects have undergone two or more conditions (i.e., the two factors are "time" and "conditions"). "

[source]

Figure 11: Repeated Measures - Factorial (Extended)



### Friedman vs Kruskal-Wallis Tests

"For comparison of three or more independent groups, Kruskal-Wallis test which is comparable to one-way ANOVA is used. Also for analysis of correlated data with three or more occasions or conditions, Friedman test, comparable to repeated measures one-way ANOVA, is used as a nonparametric method."

[source Hae-Young Kim, NCBI paper]

### Chi-Square Test vs z-test

"Mathematically identical! One is just the square of the other"

z^2 = chi-square

### McNamer's Chi Square Test

**General:** Measured X proportion at baseline and at 2 other time points for that same group.

**Example:** A group of users got a package. What proportion were active at baseline how did it change over time for that group.

***Equivalent (conditional):*** *Repeated Measures Anova (Continuous)*

### McNamer's Exact Test

We care about the **discordant pairs** (the ones that are not the same). Use the binomial distribution.

**Example:** If we have 7 discordant pairs, what is the probability of getting X = 0. $\binom{7}{0}0.5^7 0.5^0 = 0.0078$

***Equivalent (conditional):*** *Chi Square (independent)*

### Generalized Linear Models

Logistic, Poisson, Probit, Negative Binomial regression
**R Example:**
glm(y~x, family=poisson)

1. The residuals come from a distribution in the exponential family. (And yes, you need to specify which one).
2. The mean of y has a linear form with model parameters **only** through a link function. The general linear models have a simple identity link function.
3. The model parameters are estimated using Maximum Likelihood Estimation. OLS doesn't work.

[Source]

### Logistic Regression

Odds ratios can be misleading for common cases.

Not using a **Conditional Logistic Regression** when the data is correlated (such as when there are matched control/experiment pairs), would result in higher pvalue (conservative) and perhaps miss significance.

**Probit Models** will yield *similar* results. Can be generalized to account for non-constant error variances (aka heteroskedastic probit models) [MethodConsultants].

### Repeated Measures Logistic Regression

**General:** Measured X *proportion* at baseline AND at 2 or more other time points for that same group.

**Example:** A group of users got a package. What was the *proportion* of users engaged at baseline and how did it change over time.

***Equivalent (conditional):*** *Repeated Measures Anova (continuous)*

[Implementation in R]

### Linear/Logistic/Poisson/Cox Regression

Linear: Outcome = a + Bx

Logistic: Ln odds(Y) = a + Bx,
where B transformed to Odds Ratio

Poisson: Ln (count) = a + Bx,
where B transformed to Rate Ratio

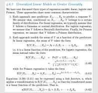
Cox: Ln (hazard) = a + Bx,
where B transformed to Hazard Ratio

### Poisson Regression

Mean = Expected Count
Variance = Mean
SD = Mean ^ 0.5

If people in a study have been following for different times, you can account for this by normalizing for their duration in study (aka incidence rate).

### Excerpt from Introduction to Statistical Learning (Edition 2) p.170



*4.6.3 Generalized Linear Models in Greater Generality*

We have now discussed three types of regression models: linear, logistic and Poisson. These approaches share some common characteristics:

1. Each approach uses predictors $X_1, \ldots, X_p$ to predict a response $Y$. We assume that, conditional on $X_1, \ldots, X_p$, $Y$ belongs to a certain family of distributions. For linear regression, we typically assume that $Y$ follows a Gaussian or normal distribution. For logistic regression, we assume that $Y$ follows a Bernoulli distribution. Finally, for Poisson regression, we assume that $Y$ follows a Poisson distribution.
2. Each approach models the mean of $Y$ as a function of the predictors. In linear regression, the mean of $Y$ takes the form

$$E(Y|X_1,\ldots,X_p) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p, \quad (4.39)$$

i.e. it is a linear function of the predictors. For logistic regression, the mean instead takes the form

$$E(Y|X_1,\ldots,X_p) = \Pr(Y=1|X_1,\ldots,X_p) = \frac{e^{\beta_0+\beta_1X_1+\cdots+\beta_pX_p}}{1+e^{\beta_0+\beta_1X_1+\cdots+\beta_pX_p}}, \quad (4.40)$$

while for Poisson regression it takes the form

$$E(Y|X_1,\ldots,X_p) = \lambda(X_1,\ldots,X_p) = e^{\beta_0+\beta_1X_1+\cdots+\beta_pX_p}. \quad (4.41)$$

Equations (4.39)–(4.41) can be expressed using a *link function*, $\eta$, which applies a transformation to $E(Y|X_1,\ldots,X_p)$ so that the transformed mean is a linear function of the predictors. That is,

$$\eta(E(Y|X_1,\ldots,X_p)) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p. \quad (4.42)$$

### Kaplan-Meir Curves

Intro to Kaplan Meir Curves

### Cox Regression

Must test the assumption of proportional hazards (the difference in hazard rates between the different weight groups must be constant).

Measures the instantaneous incident rates.

If no censoring occurs, you can use other regression techniques and if there are no other explanatory variables, a simple survival proportion works.

[additional reference paper]